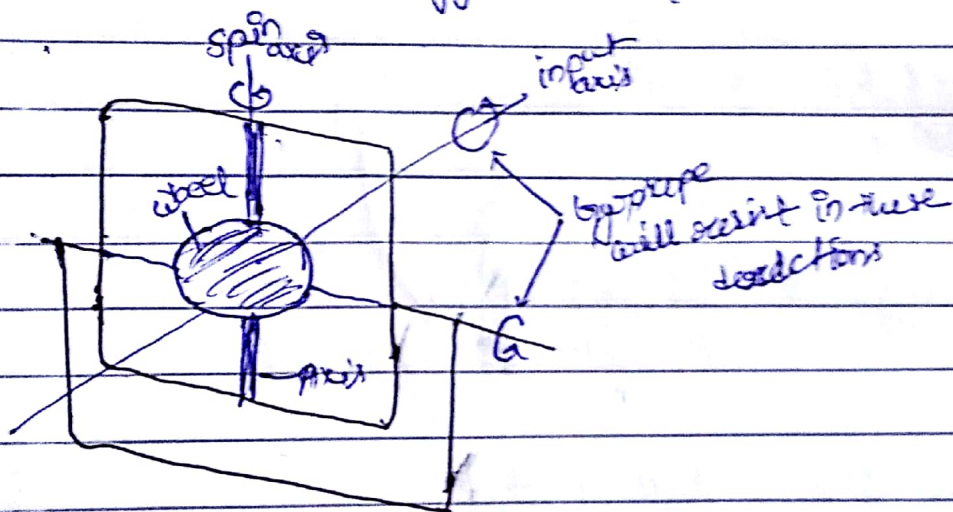


unit-2

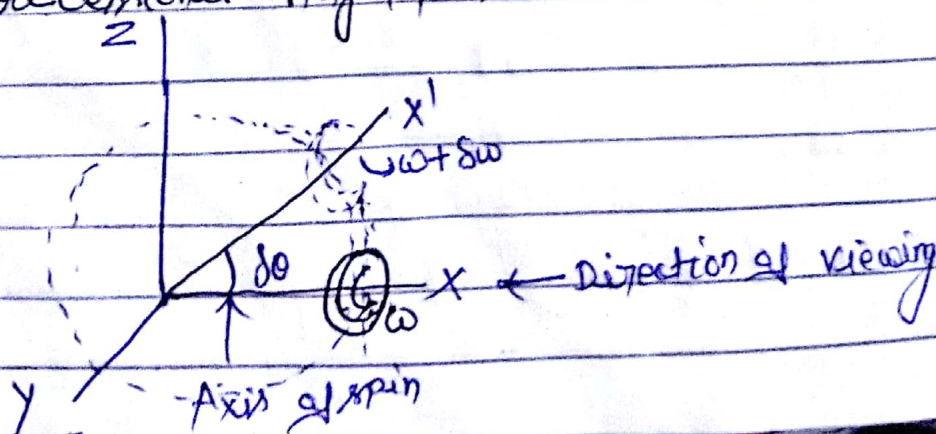
Gyroscope torque: If the axis of a spinning rotating body is given an angular motion about an axis $\perp$ to the axis of the spin, an angular acc. acts on the body about the $\perp$ axis. The torque required to produce this acc. is known as the active gyroscopic torque.

"The effect produced by the reactive gyro couple is known as the gyroscopic effect".



Note: when a body, itself is moving with angular velocity along a circular path, it is subjected to the centrifugal force radially outwards. This centrifugal force is called Reactive force.

Precessional Angular motion:-



$Ox = \text{axis of spin}$

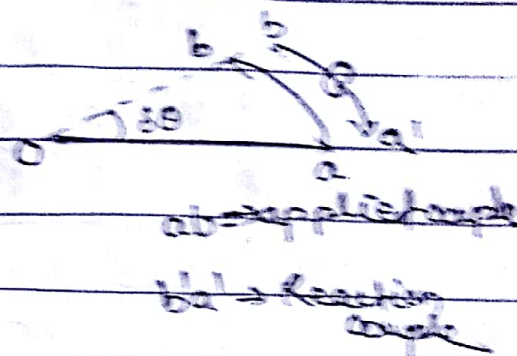
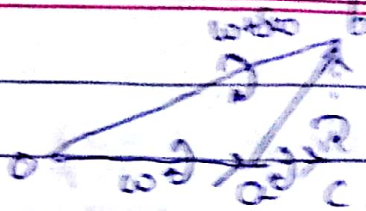
$Oy = \text{axis of gyroscopic couple}$

$Oz = \text{axis of precession}$

$yz = \text{plane of spin}$

$xy = \text{plane of precession}$

$xz = \text{plane of gyroscopic couple}$



* angular acc. = $\omega \frac{d\theta}{dt} = \omega \cdot \omega_p$

* Total angular acc. = $\frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$

Gyroscopic Torque (couple)

$ab = \omega \times \delta\theta$

$\alpha = \omega \frac{\delta\theta}{\delta t}$

$\alpha = \omega \cdot \omega_p$

$C/T = I \omega \cdot \omega_p \text{ N-m}$

Where $I = MR^2$
 $\omega = \text{angular velocity}$
 $\omega_p = \frac{d\theta}{dt}$
 angular momentum = $I\omega$

Note:- $C = m \cdot g \cdot d$
 $I = mk^2 \text{ (kg-m}^2\text{)}$
 $k = \text{radius of gyration}$

$\omega = \frac{v}{r}$

$\omega = \frac{2\pi N}{60}$

$F = \frac{C/T}{x}$

* Gyroscopic effect on naval ships:

SUTOM

$$\theta = \phi \sin \omega_0 t$$

ϕ = Amplitude of swing

ω_0 = Angular velocity of SUTOM

$$\omega_0 = \frac{2\pi}{T_p} \text{ rad/s}$$

Angular velocity of precession

$$\omega_p = \frac{d\theta}{dt} = \frac{d}{dt} (\phi \sin \omega_0 t) = \phi \omega_0 \cos \omega_0 t$$

$$C_{max} = I \cdot \omega \cdot \omega_{pmax}$$

ω_{pmax} = when $\phi \cos \omega_0 t = 1$

$$\omega_{pmax} = \phi \omega_0$$

$$\omega_{pmax} = \phi \frac{2\pi}{T_p}$$

Ex P

* Gyroscopic effect on an automobile

$$Wt. \text{ on each wheel} = \frac{W}{4} = \frac{mg}{4} \text{ (Newt)}$$

$$R_w = \frac{W}{4} = \frac{mg}{4} \text{ (Newt)}$$

gyroscopic effect due to four wheels

$$\frac{W}{C_w} = 4 \cdot I_w \cdot \omega_w \cdot \omega_p = 4 \cdot I_w \cdot \frac{v^2}{R}$$

$$\omega_w = \frac{v}{r_1}, \quad \omega_p = \frac{v}{R}$$

I_w = mass mo of each wheel

due to engine

$$\frac{C_e}{T_e} = I_e \cdot \omega_e \cdot \omega_p = I_e \cdot \phi \cdot \omega_0 \cdot \omega_p$$

$$\phi = \frac{\omega_e}{\omega_0}$$

gear ratio

$$C_g / T_g = C_w \pm C_e$$

+ve sign when the engine parts rotate in the same direction as the wheels.

-ve when engine parts rotate in the opp direction as the wheel.

$$F_{o.w} = \frac{C_g}{w} (\downarrow \text{wards})$$

O.w = outer wheels

$R_g =$

forces on two outer wheels.

$$F_{i.w} = \frac{C_g}{w} (\uparrow \text{wards})$$

(i.w = inner wheels)

forces on each of the inner wheels = $\frac{C_g}{2w} (\uparrow \text{wards})$

" " " " " " outer wheels = $\frac{C_g}{2w} (\downarrow \text{wards})$

Reaction of ground on each outer wheel $R_{g(o.w)} = \frac{C_g}{2w} (\uparrow \text{wards})$

Reaction of ground on each inner wheel $R_{g(i.w)} = \frac{C_g}{2w} (\downarrow \text{wards})$

* Effect of Centrifugal Couple

centrifugal force, $mR\omega_p^2$

$$= m \cdot R \cdot \left(\frac{v}{R}\right)^2$$

$$= m \cdot \frac{v^2}{R}$$

overturning couple

$$C_c = mR\omega_p^2 \times h = m \cdot \frac{v^2}{R} h$$

M. Imp

Gyroscopic effect on naval ship during rolling motion.

Ans: — As the axis of the rolling of the ship and of the spool are parallel, there is no ~~rotation~~ precession of the axis of spin and thus there is no gyroscopic effect.

In the same way, the effect on steering, pitching or rolling can be observed when the plane of the spin of the rotating masses is horizontal but along the longitudinal axis of the vessel as when the axis is vertical.

* Dynamic Analysis

D'Alembert principle: It states that the inertia forces and couples, and the external forces and torques on a body together give statical equilibrium.

$$\sum F + F_i = 0$$

$$F_i = -m f_g$$

f_g = acc. of centre of mass of the body.

* Dynamic Analysis of slider-crank mechanism

* Velocity of piston $v = r\omega \sin\theta$

* Acc. of piston $a = \frac{dv}{dt} = \frac{dv}{d\theta} \frac{d\theta}{dt} = \frac{d(r\omega \sin\theta)}{d\theta} \omega$

$$= r\omega^2 (\cos\theta + \frac{\cos 2\theta}{n}) \quad \text{as } n \gg \cos 2\theta$$

$$a = r\omega^2 \cos\theta$$

=

* Angular velocity of connecting rod

$$\left[\frac{L \sin\beta}{r \sin\theta} = \frac{r \cdot \omega}{r} \right]$$

$$y = L \sin\beta = r \sin\theta$$

$$\frac{L}{r} = n$$

$$\sin\beta = \frac{r \sin\theta}{L}$$

$$\cos\beta \frac{d\beta}{dt} = \frac{1}{n} \cos\theta \cdot \frac{d\theta}{dt}$$

$$\frac{d\beta}{dt} = \frac{1}{n} \frac{\cos\theta}{\cos\beta} \left(\frac{d\theta}{dt} = \omega \right)$$

$$\left(\omega_c = \omega \frac{\cos \theta}{\sqrt{1 - \sin^2 \theta}} \right)$$

$$\begin{cases} n \sin \beta = \sin \theta \\ \frac{1}{n} (1 - \cos^2 \beta) = \sin^2 \theta \\ \sin^2 \beta = \frac{\sin^2 \theta}{n^2} \\ (1 - \cos^2 \beta) = \frac{\sin^2 \theta}{n^2} \\ \cos \beta = \sqrt{1 - \frac{\sin^2 \theta}{n^2}} \end{cases}$$

$$\begin{aligned} a_c &= \frac{d^2 \omega_c}{dt^2} = \frac{d\omega_c}{d\theta} \cdot \frac{d\theta}{dt} \\ &= \omega_c^2 \frac{\sin \theta \cdot \sqrt{1 - \sin^2 \theta} - \cos \theta \cdot \left(\frac{-\sin \theta}{\sqrt{1 - \sin^2 \theta}} \right)}{(1 - \sin^2 \theta)^{3/2}} \\ &= -\omega_c^2 \sin \theta \left[\frac{n^2 - 1}{(n^2 - \sin^2 \theta)^{3/2}} \right] \end{aligned}$$

Engine Force Analysis

Imp

(i) Piston Effort (Effective driving force)

$$* \boxed{F = F_p + m g - F_b - F_f}$$

$$F_p = P_1 A - P_2 A_2$$

$$F_b = m_f \ddot{x} = m_f \omega^2 = m_f \omega^2 \left(\frac{\cos \theta}{n} \right)$$

$$F_f =$$

(ii) (Thrust) Force along connecting rod

$$* \boxed{F_c = \frac{F}{\cos \beta}}$$

(iii) Thrust on the sides of cylinder

$$* \quad F_n = F_c \sin \beta = f \tan \beta$$

(iv) Crank effort

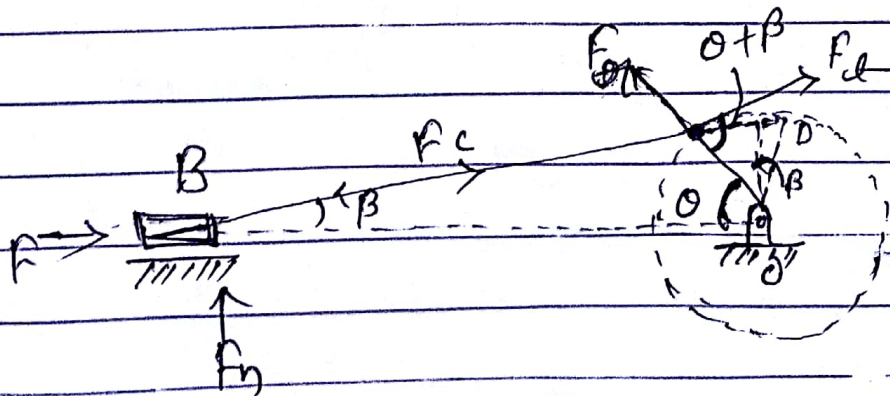
$$F_d = \frac{F \sin(\theta + \beta)}{\cos \beta}$$

(v) Thrust on the bearing

$$F_{or} = F_c \cos(\theta + \beta) = \frac{F \cos(\theta + \beta)}{\cos \beta}$$

(vi) Turning moment on crank shaft

$$\begin{aligned} T &= F_d \times r \\ &= \frac{F \sin(\theta + \beta) \times r}{\cos \beta} \\ &= \end{aligned}$$



$$* \quad \eta = \frac{l}{g} \quad (\text{oblique ratio})$$

CAMS

"A cam is a mechanical member used to impart desired motion to a follower by direct contact".

The cam may be rotating or reciprocating whereas the follower may be rotating, reciprocating or oscillating.

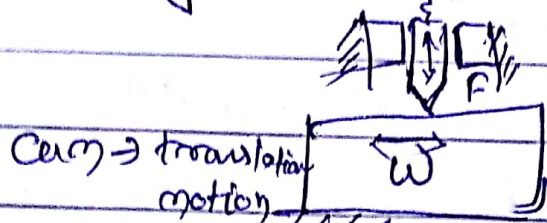
Classification

Types of cam

- 1) Shape
- 2) Follower movement
- 3) manner of constraint of the follower

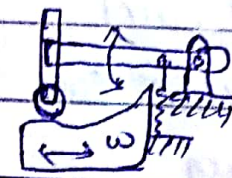
1) According to shape:-

(i) Wedge and Flat cams:

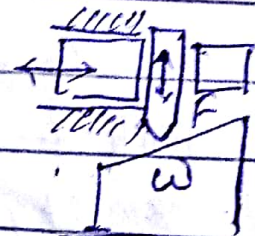


Cam $\rightarrow$ translation motion

Follower $\rightarrow$ translation

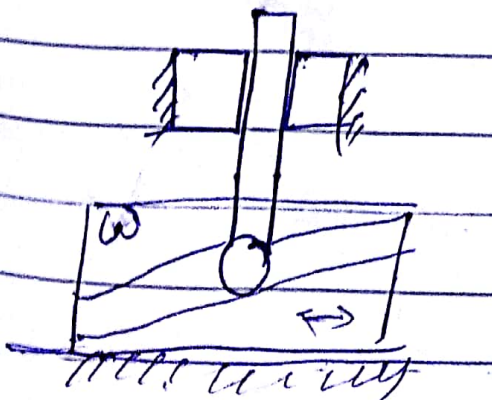


Follower $\rightarrow$ rotational wedge cam motion



Follower constraint of guide causes the relative motion.

flat cam



(ii) Radial or Disc Cam

(iii) Spiral Cam

(iv) Cylindrical Cam

(v) Conjugate Cam

(vi) Grooved Cam

(vii) Spherical Cam

2) Acc. to follower movement

i) Rise-Return-Rise (R-R-R)

ii) Dwell-Rise-Return-Dwell (D-R-R-D)

iii) Dwell-Rise-Dwell-Return-Dwell (D-R-D-R-D)

3) Acc. to constraint of the follower

i) Force-loaded cam

ii) Positive-drive cam

iii) Gravity cam

Types of followers

1) Shape

i) knife-edge follower

ii) Roller follower

iii) Mushroom follower

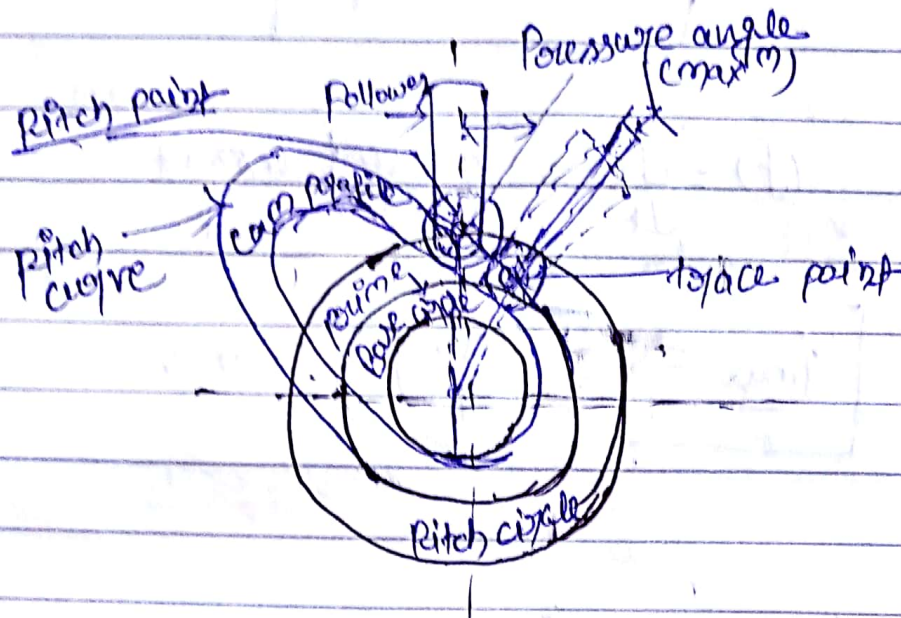
2) Movement

i) Reciprocating ~~movement~~ follower

ii) oscillating follower

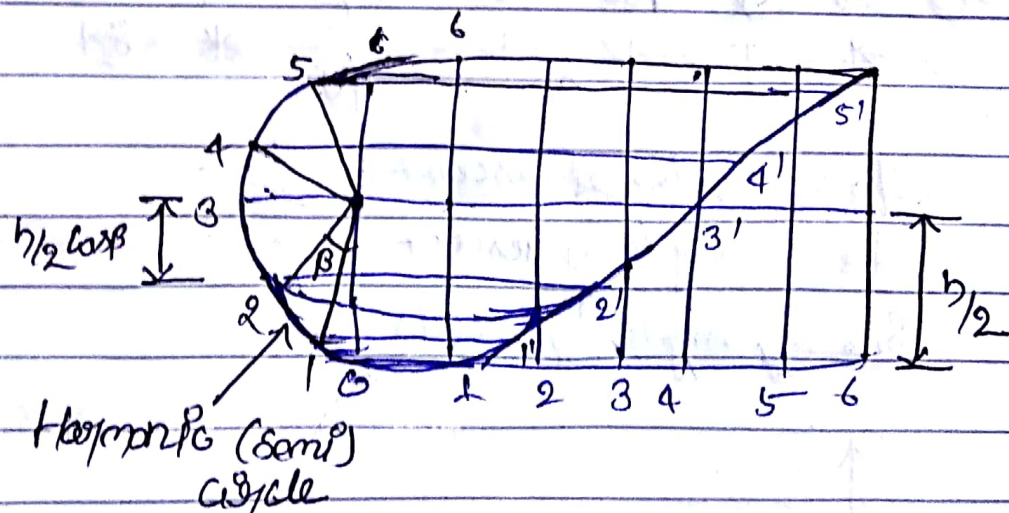
3) According to location of line of movement

- i) Radial follower
- ii) Offset follower



Motions of the follower

20.50.10.11



- S = follower displacement (instantaneous)
 h = maximum follower displacement
 v = velocity of the follower
 a = acc. of the follower
 θ = cam rotation angle (instantaneous)
- ϕ = cam rotation angle for h
 β = angle of harmonic circle

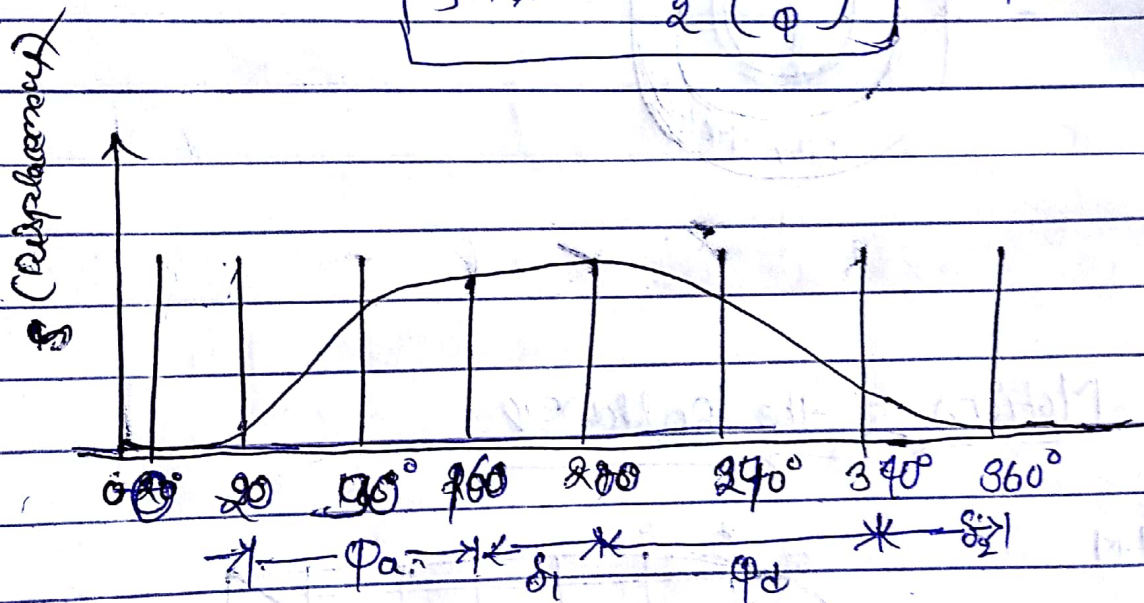
* Velocity (~~amplitude~~) $v = \frac{ds}{dt} = \frac{h}{2} \cdot \pi \omega \sin \pi \omega t$

$v_{\max} = \text{when } 2\theta = \pi$

$$v_{\max} = \frac{h}{2} \frac{\pi \omega}{\phi}$$

* Acceleration (f) $= \frac{dv}{dt} = \frac{h}{2} \left(\frac{\pi \omega^2}{\phi} \right) \cos \pi \omega t$

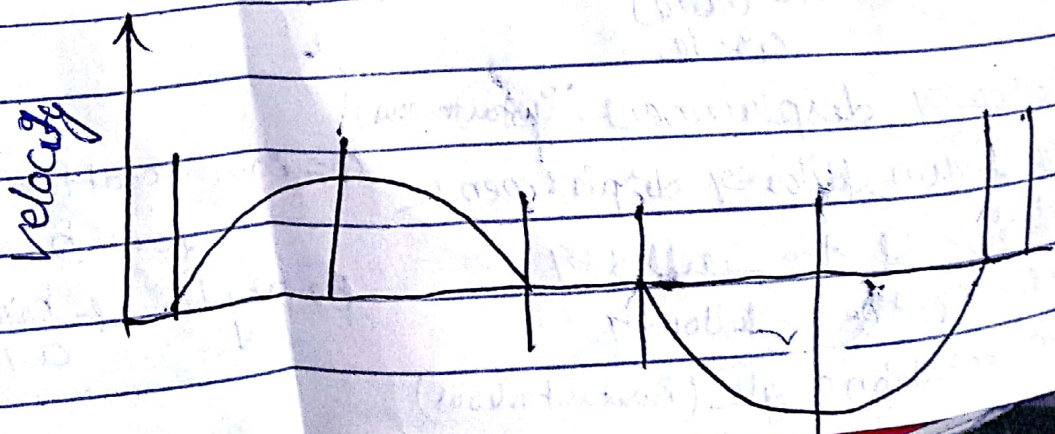
$$f_{\max} = \frac{h}{2} \left(\frac{\pi \omega^2}{\phi} \right) \text{ at } \theta = 0$$



$\phi_a = \text{angle of ascent}$

$\phi_d = \text{angle of descent}$

$\delta_{1,2} = \text{Angle of dwells.}$



acceleration

jerk

Note: ϕ_a = angle of ascent : it is the angle through which the cam turns while the follower rises

ϕ_d = angle of descent : it is the angle through which cam turns during the time the follower returns to its initial position

ϕ_s = angle of dwell : it is the angle through which the cam turns during follower remains stationary at highest or the lowest position

* kinematic derivative

(displacement) $s = s(\theta)$

(velocity) $\dot{s}(\theta) = \frac{ds}{d\theta}$

(acc) $\ddot{s}(\theta) = \frac{d^2s}{d\theta^2}$

(jerk) $\dddot{s}(\theta) = \frac{d^3s}{d\theta^3}$